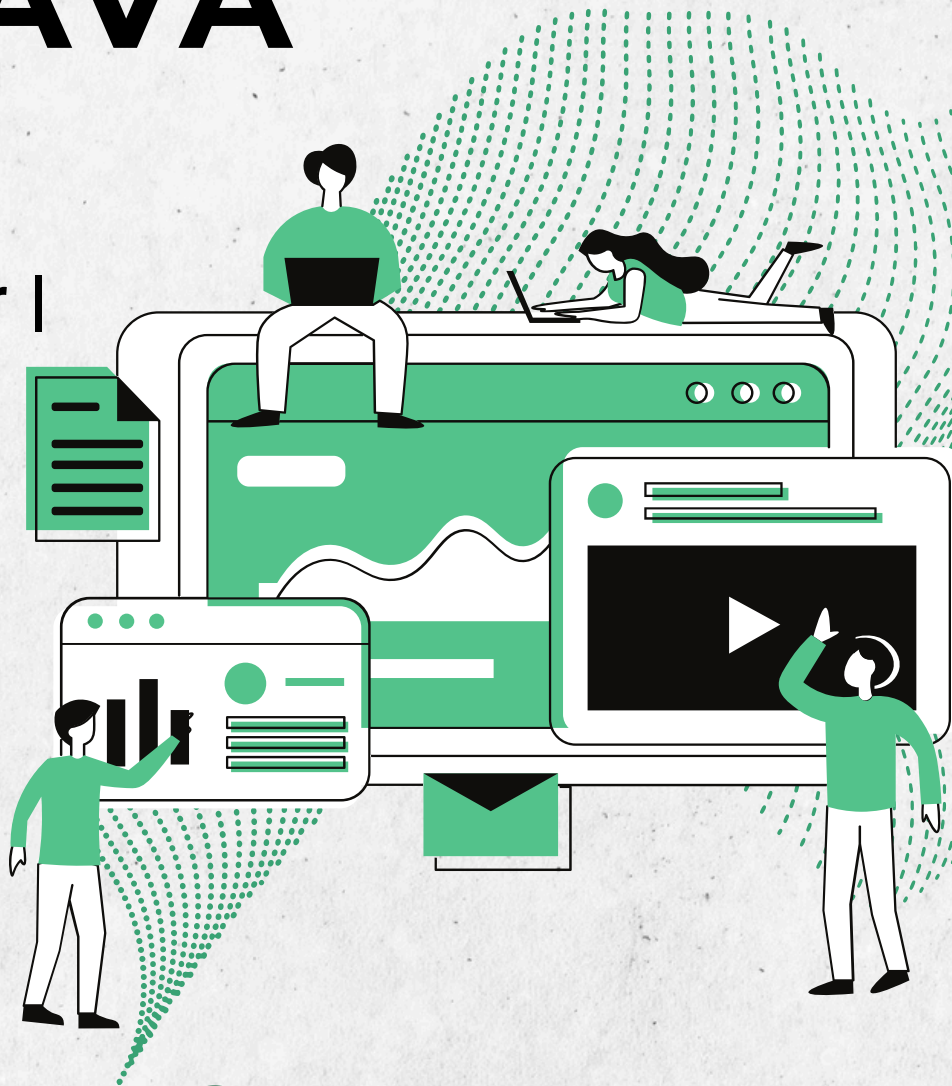




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# ANNOTATIONS IN SPRING BOOT AND JAVA

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# JAVA + SPRING BOOT ANNOTATIONS MASTER CHART (100+)

| #  | Annotation               | Layer / Category | Purpose / Use                                                                              |
|----|--------------------------|------------------|--------------------------------------------------------------------------------------------|
| 1  | @SpringBootApplication   | Configuration    | Combines @Configuration, @EnableAutoConfiguration, @ComponentScan to start Spring Boot app |
| 2  | @Configuration           | Configuration    | Declares that a class contains bean definitions                                            |
| 3  | @ComponentScan           | Configuration    | Scans the package for Spring-managed components                                            |
| 4  | @EnableAutoConfiguration | Configuration    | Enables auto-configuration of beans based on classpath                                     |
| 5  | @Bean                    | Configuration    | Declares a method that returns a Spring-managed bean                                       |
| 6  | @PropertySource          | Configuration    | Loads external properties file into Spring Environment                                     |
| 7  | @Import                  | Configuration    | Imports additional configuration classes                                                   |
| 8  | @Value                   | Configuration    | Injects values from properties or environment                                              |
| 9  | @Profile                 | Configuration    | Activates beans/config based on active profile                                             |
| 10 | @DependsOn               | Configuration    | Specifies bean initialization order                                                        |

# DEPENDENCY INJECTION & BEAN MANAGEMENT

| #  | Annotation      | Layer / Category | Purpose / Use                                            |
|----|-----------------|------------------|----------------------------------------------------------|
| 11 | @Autowired      | Core / Bean      | Injects dependencies automatically by type               |
| 12 | @Qualifier      | Core / Bean      | Resolves conflict when multiple beans of same type exist |
| 13 | @Primary        | Core / Bean      | Marks a bean as the default choice when multiple exist   |
| 14 | @Lazy           | Core / Bean      | Delays bean initialization until needed                  |
| 15 | @Scope          | Core / Bean      | Defines bean scope: singleton, prototype, etc.           |
| 16 | @Component      | Core / Bean      | Marks a class as a Spring-managed component              |
| 17 | @Service        | Service Layer    | Marks a class as a business service component            |
| 18 | @Repository     | DAO / Repository | Marks a class as a DAO and enables exception translation |
| 19 | @Controller     | Web Layer        | Marks a web controller handling MVC requests             |
| 20 | @RestController | Web Layer        | Combines @Controller + @ResponseBody for REST APIs       |

## WEB & REST CONTROLLER LAYER

| #  | Annotation            | Layer      | Purpose / Use                                           |
|----|-----------------------|------------|---------------------------------------------------------|
| 21 | @RequestMapping       | Controller | Maps HTTP requests to handler methods/classes           |
| 22 | @GetMapping           | Controller | Shortcut for GET request                                |
| 23 | @PostMapping          | Controller | Shortcut for POST request                               |
| 24 | @PutMapping           | Controller | Shortcut for PUT request                                |
| 25 | @DeleteMapping        | Controller | Shortcut for DELETE request                             |
| 26 | @PatchMapping         | Controller | Shortcut for PATCH request                              |
| 27 | @RequestParam         | Controller | Binds request parameters to method arguments            |
| 28 | @PathVariable         | Controller | Binds URI template variables to method parameters       |
| 29 | @RequestBody          | Controller | Binds HTTP request body to a Java object                |
| 30 | @ResponseBody         | Controller | Indicates method return is written directly to response |
| 31 | @ResponseStatus       | Controller | Sets custom HTTP status for a method                    |
| 32 | @CrossOrigin          | Controller | Enables Cross-Origin Resource Sharing (CORS)            |
| 33 | @ExceptionHandler     | Controller | Handles specific exceptions in controllers              |
| 34 | @ControllerAdvice     | Controller | Global exception handling across controllers            |
| 35 | @RestControllerAdvice | Controller | Same as above but for REST APIs                         |

# PERSISTENCE / JPA / HIBERNATE ANNOTATIONS

| #  | Annotation      | Layer | Purpose / Use                            |
|----|-----------------|-------|------------------------------------------|
| 36 | @Entity         | JPA   | Marks a class as a JPA entity            |
| 37 | @Table          | JPA   | Specifies table name and details         |
| 38 | @Id             | JPA   | Marks a field as primary key             |
| 39 | @GeneratedValue | JPA   | Defines auto-generation strategy for IDs |
| 40 | @Column         | JPA   | Customizes column name, length, etc.     |
| 41 | @Transient      | JPA   | Field not persisted in database          |
| 42 | @Lob            | JPA   | Large Object (BLOB/CLOB)                 |
| 43 | @Enumerated     | JPA   | Maps enum to database column             |
| 44 | @OneToOne       | JPA   | One-to-one relationship                  |
| 45 | @OneToMany      | JPA   | One-to-many relationship                 |
| 46 | @ManyToOne      | JPA   | Many-to-one relationship                 |
| 47 | @ManyToMany     | JPA   | Many-to-many relationship                |
| 48 | @JoinColumn     | JPA   | Defines foreign key column               |
| 49 | @JoinTable      | JPA   | Defines join table for relationships     |
| 50 | @Embedded       | JPA   | Embeds another class inside entity       |
| 51 | @Embeddable     | JPA   | Marks class that can be embedded         |
| 52 | @Version        | JPA   | For optimistic locking                   |

## TRANSACTION & AOP LAYER

| #  | Annotation                   | Layer        | Purpose / Use                        |
|----|------------------------------|--------------|--------------------------------------|
| 53 | @Transactional               | Service / DB | Declares a transactional boundary    |
| 54 | @EnableTransactionManagement | Config       | Enables @Transactional processing    |
| 55 | @Aspect                      | AOP          | Declares a class as an aspect        |
| 56 | @Before                      | AOP          | Advice before method execution       |
| 57 | @After                       | AOP          | Advice after method execution        |
| 58 | @AfterReturning              | AOP          | Runs after successful execution      |
| 59 | @AfterThrowing               | AOP          | Runs after exception is thrown       |
| 60 | @Around                      | AOP          | Runs before and after target method  |
| 61 | @Pointcut                    | AOP          | Defines reusable pointcut expression |

# VALIDATION ANNOTATIONS (JAKARTA / HIBERNATE VALIDATOR)

| #  | Annotation | Layer      | Purpose / Use                       |
|----|------------|------------|-------------------------------------|
| 62 | @Valid     | Validation | Triggers validation on request body |
| 63 | @NotNull   | Validation | Ensures value is not null           |
| 64 | @NotEmpty  | Validation | Ensures value is not null or empty  |
| 65 | @NotBlank  | Validation | Ensures String is not blank         |
| 66 | @Min       | Validation | Minimum value for number            |
| 67 | @Max       | Validation | Maximum value for number            |
| 68 | @Email     | Validation | Validates email format              |
| 69 | @Pattern   | Validation | Validates regex pattern             |
| 70 | @Size      | Validation | Ensures size within limits          |
| 71 | @Past      | Validation | Validates past date                 |
| 72 | @Future    | Validation | Validates future date               |

## LOMBOK ANNOTATIONS (REDUCE BOILERPLATE)

| #  | Annotation          | Layer   | Purpose / Use                                       |
|----|---------------------|---------|-----------------------------------------------------|
| 73 | @Getter             | Utility | Generates getter methods                            |
| 74 | @Setter             | Utility | Generates setter methods                            |
| 75 | @Data               | Utility | Combines getter, setter, toString, equals, hashCode |
| 76 | @NoArgsConstructor  | Utility | Creates no-arg constructor                          |
| 77 | @AllArgsConstructor | Utility | Creates constructor with all fields                 |
| 78 | @Builder            | Utility | Implements builder pattern                          |
| 79 | @ToString           | Utility | Generates toString()                                |
| 80 | @EqualsAndHashCode  | Utility | Generates equals() and hashCode()                   |



## SCHEDULING & ASYNC PROCESSING

| #  | Annotation        | Layer   | Purpose / Use                      |
|----|-------------------|---------|------------------------------------|
| 81 | @EnableScheduling | Config  | Enables scheduled tasks            |
| 82 | @Scheduled        | Service | Executes method at fixed rate/cron |
| 83 | @EnableAsync      | Config  | Enables async method execution     |
| 84 | @Async            | Service | Executes method asynchronously     |

## SECURITY (SPRING SECURITY)

| #  | Annotation                  | Layer    | Purpose / Use                      |
|----|-----------------------------|----------|------------------------------------|
| 85 | @EnableWebSecurity          | Config   | Enables web security config        |
| 86 | @PreAuthorize               | Security | Checks authorization before method |
| 87 | @PostAuthorize              | Security | Checks authorization after method  |
| 88 | @Secured                    | Security | Restricts access to roles          |
| 89 | @RolesAllowed               | Security | Restricts access to allowed roles  |
| 90 | @AuthenticationPrincipal    | Security | Injects current user principal     |
| 91 | @EnableGlobalMethodSecurity | Security | Enables method-level security      |

# TESTING ANNOTATIONS

| #   | Annotation                          | Layer | Purpose / Use                                |
|-----|-------------------------------------|-------|----------------------------------------------|
| 92  | @SpringBootTest                     | Test  | Loads full app context for integration tests |
| 93  | @MockBean                           | Test  | Creates mock bean in Spring context          |
| 94  | @DataJpaTest                        | Test  | Tests JPA repositories only                  |
| 95  | @WebMvcTest                         | Test  | Tests only the web layer                     |
| 96  | @AutoConfigureMockMvc               | Test  | Configures MockMvc instance                  |
| 97  | @ExtendWith(MockitoExtension.class) | Test  | Integrates Mockito with JUnit 5              |
| 98  | @BeforeEach                         | Test  | Runs before each test                        |
| 99  | @AfterEach                          | Test  | Runs after each test                         |
| 100 | @Test                               | Test  | Marks a test method                          |

## OTHER USEFUL / RARE ANNOTATIONS

| #   | Annotation           | Category  | Purpose / Use                                     |
|-----|----------------------|-----------|---------------------------------------------------|
| 101 | @Conditional         | Config    | Enables bean conditionally                        |
| 102 | @Order               | Core      | Defines execution order for components            |
| 103 | @Deprecated          | Java Core | Marks method/class as deprecated                  |
| 104 | @Override            | Java Core | Indicates method overrides parent method          |
| 105 | @FunctionalInterface | Java Core | Marks interface as functional (1 abstract method) |
| 106 | @SafeVarargs         | Java Core | Suppresses heap pollution warnings                |
| 107 | @SuppressWarnings    | Java Core | Suppresses compiler warnings                      |
| 108 | @Record              | Java Core | Marks compact immutable data class (Java 16+)     |